



PROVINCIAL EXAMINATION
NOVEMBER 2022
GRADE 11
MARKING GUIDELINES

MATHEMATICS (PAPER 2)

22 pages

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INSTRUCTIONS AND INFORMATION

- **A** – ACCURACY
- **CA** – CONSISTENT ACCURACY
- **S** – STATEMENT
- **R** – REASON
- **S/R** – STATEMENT with REASON

NOTES:

- If a candidate answered a question TWICE, mark only the first attempt.
- If a candidate crossed OUT an answer and did not redo it, mark the crossed-out answer.
- Consistent accuracy applies to ALL aspects of the marking guidelines.
- Assuming values/answers in order to solve a question is UNACCEPTABLE.

QUESTION 1

1.1		✓ Box (Q1 and Q3) ✓ Whiskers (Min. and Max.) ✓ Median	(3)
1.2	The data is distributed symmetrically, a normal distribution. OR The data is slightly skewed to the right.	✓ Explanation	(1)
1.3	$\sigma = 23,77$	✓✓ σ	(2)
1.4	$(\bar{x} - \sigma; \bar{x} + \sigma)$ $= (287 - 23,77; 287 + 23,77)$ $= (263,23; 310,77)$ $\therefore$ 6 municipalities are within one standard deviation.	✓ $\bar{x} = 287$ ✓ Interval ✓ 6	(3)
1.5	$S-IQR = \frac{1}{2} (Q_3 - Q_1)$ $= \frac{1}{2} (306 - 266)$ $= 20$	✓ $Q_3 - Q_1$ ✓ $\frac{1}{2} \times$ ✓ Answer	(3)
			[12]

Commented [BK1]: Wrong values on Box and Whisker

QUESTION 2

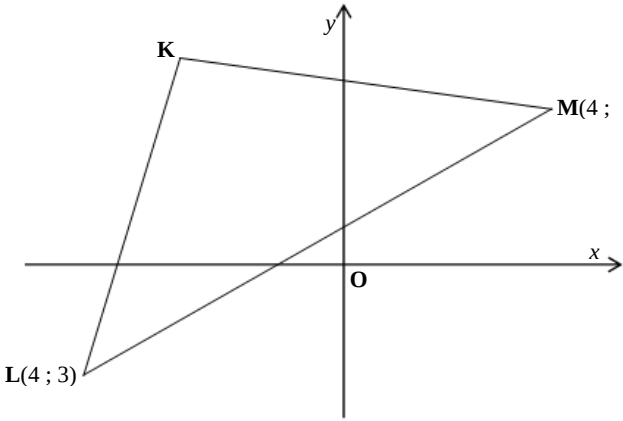
2.1	$p = 5$ $q = 27$ $r = 1$	$\checkmark p$ $\checkmark q$ $\checkmark r$	(3)
2.2	CUMULATIVE FREQUENCY GRAPH (OGIVE)		
2.3	$M = 46$	$\checkmark\checkmark M = 46(\pm 2)$	(2)
2.4	$32 - 24$ $= 8$ $\therefore 8$ learners achieved a mark of greater than 55.	$\checkmark CF = 24(\pm 1)$ $\checkmark 32 - CF$	(2)
			[10]

QUESTION 3

3.1	$AB = \sqrt{(0-10)^2 + (7-2)^2}$ $AB = \sqrt{125}$ $AB = 5\sqrt{5}$	✓ Substitute into correct formula ✓ AB ✓ Simplified surd	(3)
3.2	$m_{AC} = \frac{7-(-4)}{0-7}$ $m_{AC} = -\frac{11}{7}$	✓ Gradient substitute ✓ m_{AC}	(2)
3.3	$\tan \theta = -\frac{11}{7}$ $RA = 57,528 \dots^\circ$ $\theta = 180^\circ - 57,528 \dots^\circ$ $\theta = 122,47^\circ$	✓ Angle of inclination substitute (CA 4.2) ✓ RA ✓ θ (subtract RA from 180°)	(3)

3.4	$\tan \alpha = 2$ $\alpha = 63,434 \dots^\circ$ $\hat{A}CB = 122,47^\circ - 63,434 \dots^\circ$ $\hat{A}CB = 59,04^\circ$ $m_{AB} = \frac{1}{2}$ $\hat{C}AB = 180^\circ - \tan^{-1}\left(\frac{1}{2}\right) - 122,74^\circ$ $\hat{C}AB = 30,69^\circ$ $\hat{A}CD = 30,69^\circ$ [alt $\angle$'s, $AB \parallel CD$] $\hat{BCD} = 30,69^\circ + 59\ 04^\circ$ $\hat{BCD} = 89,73^\circ$ OR IF θ is not rounded $\tan \alpha = 2$ $\alpha = 63,434 \dots^\circ$ $\hat{A}CB = 122,4711 \dots^\circ - 63,434 \dots^\circ$ $\hat{A}CB = 59,036 \dots^\circ$ $m_{AB} = \frac{1}{2}$ $\hat{C}AB = 180^\circ - \tan^{-1}\left(\frac{1}{2}\right) - 122,74 \dots^\circ$ $\hat{C}AB = 30,963^\circ$ $\hat{A}CD = 30,963^\circ$ [alt $\angle$'s, $AB \parallel CD$] $\hat{BCD} = 30,963^\circ + 59\ 036 \dots^\circ$ $\hat{BCD} = 90^\circ$	α $\hat{A}CB$ $\hat{C}AB$ $\hat{BCD}$ OR $\checkmark \alpha$ $\checkmark \hat{A}CB$ $\checkmark \hat{C}AB$ $\checkmark \hat{BCD}$	(4)
			[12]

QUESTION 4

			
4.1	$y = 5x + 9$ and $5y + x - 19 = 0$ $5(5x + 9) + x = 19$ $26x = -26$ $x = -1$ $y = 5(-1) + 9$ $y = 5$ $\therefore K(-1; 4)$	✓ Sub y ✓ Solve for x ✓ Sub x back NO MARK for coordinates (given)	(3)
4.2	$m_{KL} = 5$ and $m_{KM} = -\frac{1}{5}$ $\therefore m_{KL} \times m_{KM} = -1$ $KL \perp KM$ $\therefore \angle LKM = 90^\circ$	✓ m_{KL} ✓ m_{KM} ✓ Definition of perpendicular gradients	(3)

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4.3	$KL = \sqrt{((-1 - (-2))^2 + (4 - (-1))^2)}$ $KL = \sqrt{26}$ $KN = \sqrt{(-1 - 4)^2 + (4 - 3)^2}$ $KN = \sqrt{26}$ $\text{Area } \Delta KML = \frac{1}{2} (\sqrt{26})(\sqrt{26})$ $\text{Area } \Delta KML = 13 \text{ units}^2$	✓ KL length ✓ KN length ✓ Sub Area formula ✓ Area ΔKML	(4)
4.4	$M \left(\frac{-2+4}{2}; \frac{3+(-1)}{2} \right)$ $M(1; 1)$ $m_{NL} = \frac{3 - (-1)}{4 - (-2)}$ $m_{NL} = \frac{2}{3}$ $y - y_1 = -\frac{2}{3}(x - x_1)$ $y - 1 = -\frac{2}{3}(x - 1)$ $y = -\frac{2}{3}x + \frac{5}{2}$	✓ Midpoint of NL ✓ Gradient of NL ✓ Perpendicular gradient ✓ Substitute Midpoint ✓ Equation	(5)
4.5	$m_{NL} = m_{NP} = m_{LP}$ $\frac{2}{3} = \frac{y-3}{7-4}$ $y = 5$	✓ Definition of collinear ✓ Sub P(7; y)	(2)

Commented [BK2]: KL not ML

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4.6	Q(3 ; -2)	✓ x_Q ✓ y_Q	(2)
4.7	$\hat{K} = 90^\circ$ [proven in 4.1] $KL = KN = \sqrt{26}$ [proven in 4.2] $\therefore$ KLQN is a square [a parallelogram with an interior $\angle$ of 90° and adjacent sides equal]	✓ Interior angle of 90° ✓ Adjacent sides equal	(2)
			[21]

QUESTION 5

5.1	$\frac{\tan(180^\circ - x) \cos(360^\circ + x)}{2 \cos(90^\circ + x)}$ $= \frac{-\tan x \cdot \cos x}{-2 \sin x}$ $= \frac{-\frac{\sin x}{\cos x} \cdot \cos x}{-2 \sin x}$ $= \frac{-\sin x}{-2 \sin x}$ $= \frac{1}{2}$		✓ $-\tan x$ ✓ $\cos x$ ✓ $-2 \sin x$ ✓ $-\tan x$ identity ✓ simplify ✓ answer	(6)
5.2	5.2.1	$\text{LHS} = \frac{\sin \theta - 2 \sin \theta \cos \theta}{2 \cos^2 \theta + \cos \theta - 1}$ $= \frac{\sin \theta (1 - 2 \cos \theta)}{(2 \cos \theta - 1)(\cos \theta + 1)}$ $= \frac{-\sin \theta (2 \cos \theta - 1)}{(2 \cos \theta - 1)(\cos \theta + 1)}$ $= \frac{-\sin \theta}{\cos \theta + 1}$ $= \text{RHS}$	✓ $\sin \theta (1 - 2 \cos \theta)$ ✓ $(2 \cos \theta - 1)(\cos \theta + 1)$ ✓ Sign change/negative factor ✓ Simplify <u>with</u> conclusion	(4)

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5.2.2	Undefined when $2\cos\theta - 1 = 0$ $\cos\theta = \frac{1}{2}$ $RA = 60^\circ$ QI : $\theta = 60^\circ + k.360^\circ ; k \in Z$ QIV : $\theta = 300^\circ + k.360^\circ$ OR when $\cos\theta = -1$ $\theta = 180^\circ + k.360^\circ ; k \in Z$ NOTE: Accept other equivalent general solutions. $\cos\theta = -1$ $\theta = 180^\circ + k.360^\circ ; k \in Z$	✓ $2\cos\theta - 1 = 0$ ✓ QI general solution ✓ QII general solution ✓ $k \in Z$ ✓ $\cos\theta = -1$ general solution	
			(5)

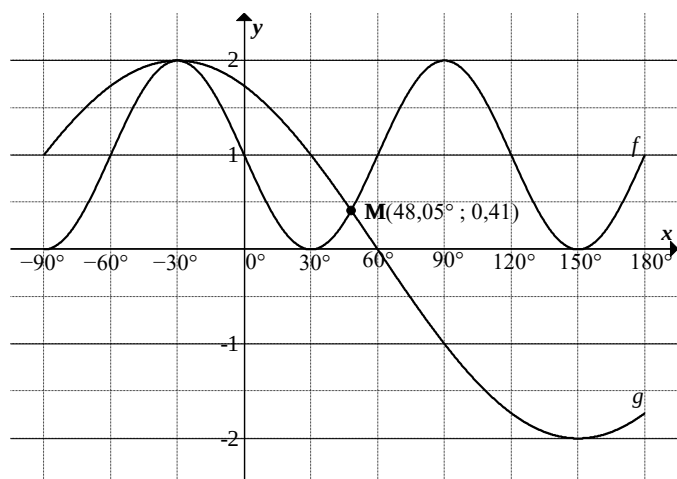
Commented [BK3]: QIV not QII

Commented [BK4]: K element of integers Z

Commented [BK5]: equation

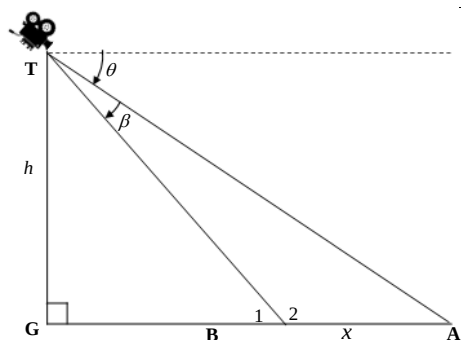
5.3				
	5.3.1	$b^2 + 1^2 = 4^2$ $b = \pm\sqrt{15}$ $b = -\sqrt{15}$ (QII)	✓ Pythagoras substitution ✓ $b = -\sqrt{15}$	(2)
	5.3.2	(a) $\tan \alpha = -\frac{1}{\sqrt{15}}$	✓ $\frac{y}{x}$ ✓ CA x	(2)
		(b) $\sin(-\alpha - 180^\circ) + \cos(-\alpha)$ $= \sin \alpha + \cos \alpha$ $= \left(\frac{1}{4}\right) + \left(\frac{-\sqrt{15}}{4}\right)$ $= \left(\frac{1 - \sqrt{15}}{4}\right)$	✓ $\sin \alpha$ ✓ $\cos \alpha$ ✓ Substitute x ; y and r	(3)

QUESTION 6



6.1	$A = 2$	✓ Answer	(1)
6.2	120°	✓ Answer	(1)
6.3	$a = -1$ $b = 3$ $p = 30^\circ$	✓ a ✓ b ✓ p	(3)
6.4	6.4.1 $-90^\circ \leq x < 60^\circ$	✓ Interval	(1)
	6.4.2 $x = -30^\circ$ or $48,05^\circ \leq x \leq 180^\circ$	✓ Interval ✓ $x = -30^\circ$	(2)
	6.4.3 $60^\circ < x < 150^\circ$ or $150^\circ < x \leq 180^\circ$	✓ 60° and 180° ✓ Not incl. 150° but incl. 180°	(2)
6.5	$t = -30^\circ + k \cdot 360^\circ ; k \in \mathbb{Z}$	✓✓ $t = -30^\circ$ ✓ $k \cdot 360^\circ$	(3)
			[13]

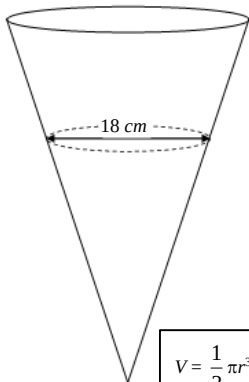
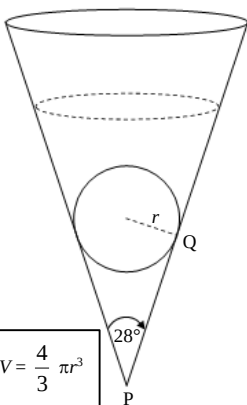
QUESTION 7



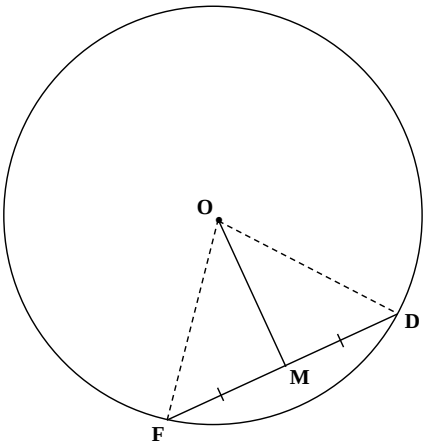
7.1	$\hat{B}_1 = \theta + \beta$	✓ Answer	(1)
7.2	In $\triangle TBG$ $\sin(\theta + \beta) = \frac{h}{BT}$ $BT = \frac{h}{\sin(\theta + \beta)}$ In $\triangle ABT$ $\hat{A} = \theta$ $\frac{x}{\sin \beta} = \frac{BT}{\sin \theta}$ $\frac{x}{\sin \beta} = \frac{h}{\sin(\theta + \beta) \sin \theta}$ $x = \frac{h \sin \beta}{\sin(\theta + \beta) \sin \theta}$	✓ Correct ratio in $\triangle TBG$ ✓ BT ✓ Sine rule substitution ✓ BT substitution ✓ simplification	(5)
7.3	$x = \frac{h \sin \beta}{\sin(\theta + \beta) \sin \theta}$ $x = \frac{20 \sin 37^\circ}{\sin(65^\circ + 37^\circ) \sin 65^\circ}$ $x = 13,58 \text{ m}$ The cat travelled 13,58 m	✓ Substitution ✓ Answer	(2)
			[8]

Commented [BK6]: Not = but +

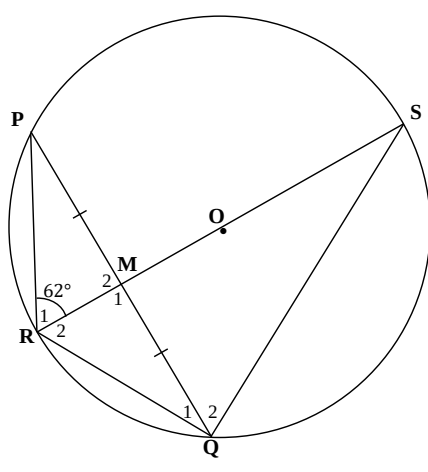
QUESTION 8

FIGURE A		FIGURE B	
			
$V = \frac{1}{3} \pi r^3$ or $V = \frac{4}{3} \pi r^3$			
8.1	$V = \frac{1}{3} \pi (9)^3$ $V = 763,41 \text{ cm}^3$	✓ Substitution ($r = 9$) ✓ Answer	(2)
8.2	$V = \frac{4}{3} \pi r^3$ $268,08 = \frac{4}{3} \pi r^3$ $\sqrt[3]{\frac{3 \times 268,08}{4\pi}} = r$ $r = 4$ $\tan 14^\circ = \frac{4}{PQ}$ $PQ = \frac{4}{\tan 14^\circ}$ $PQ = 16,04 \text{ cm}$	✓ Substitute volume into correct formula ✓ Subject of the formula ✓ r ✓ $\tan 14^\circ$ ✓ PQ	(5)
		[7]	

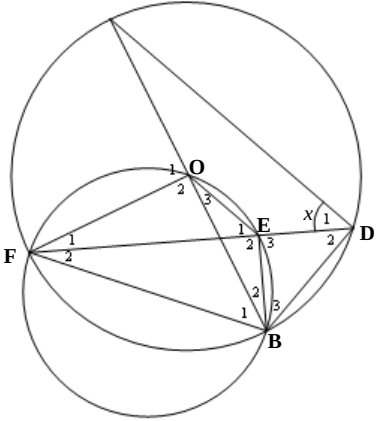
QUESTION 9

9.1	 <i>Construction: Join OF and OD</i> In $\triangle OFM$ and $\triangle ODM$ i. $FM = MD$ [GIVEN] ii. OM is a common side iii. $OF = OD$ [Radii] $\therefore \triangle OFM \cong \triangle ODM$ [SSS] $\hat{M}_1 = \hat{M}_2$ [$\triangle OFM \cong \triangle ODM$] But $\hat{M}_1 + \hat{M}_2 = 180^\circ$ [$\angle$'s on a straight line] $\therefore \hat{M}_1 = \hat{M}_2 = 90^\circ$ $\therefore OM \perp FD$	✓ Construction ✓ R ✓ R ✓ S (4)
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Commented [BK7]: + not =

9.2			
9.2.1	$\hat{M}_2 = 90^\circ$ [line from centre to midpoint of chord]	✓ S ✓ R	(2)
9.2.2	$\hat{Q}_1 + \hat{Q}_2 = 90^\circ$ [$\angle$'s in a semi-circle]	✓ S ✓ R	(2)
9.2.3	$\hat{P} = 28^\circ$ [int $\angle$'s of a Δ]	✓ S/R	
	$\hat{S} = \hat{P} = 28^\circ$ [$\angle$'s in the same segment]	✓ R	(2)
			[10]

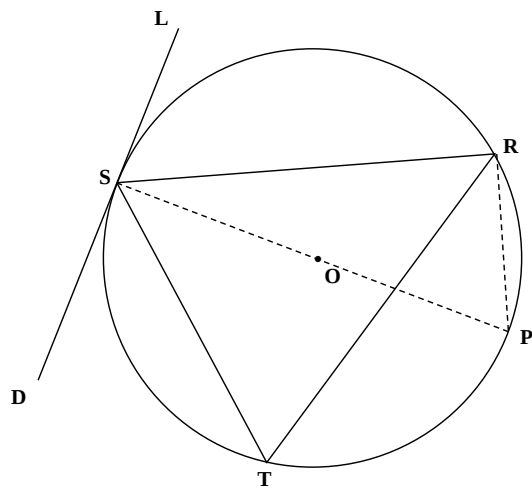
QUESTION 10

				
10.1	$\hat{B}_1 = \hat{D}_1 = x$ [∠'s in the same segment] $\hat{B}_1 = \hat{E}_1 = x$ [∠'s in the same segment]	✓ S ✓ S	(2)	
10.2	10.2.1 $\hat{O}_1 = 2x$ [∠ at centre = $2 \times$ ∠ at the circumf]	✓ S ✓ R	(2)	
	10.2.2 $\hat{O}_1 = \hat{OFB} + \hat{B}_1$ [ext ∠ of Δ] $2x = \hat{OFB} + x$ $\hat{OFB} = x$	✓ R ✓ $\hat{OFB} = x$	(2)	
	10.2.3 $\hat{D}_2 = 90^\circ - x$ [∠'s in a semi circle]	✓ S ✓ R	(2)	
	10.2.4 $\hat{O}_2 = 180^\circ - 2x$ [∠'s on a str line] $\hat{O}_2 = \hat{E}_2 = 180^\circ - 2x$ [∠'s in the same segment]	✓ S ✓ R ✓ S ✓ R	(4)	
10.3	$\hat{E}_2 = \hat{B}_3 + \hat{D}_2$ [ext ∠ of Δ] $\therefore \hat{B}_3 = 90^\circ - x$ $\therefore \hat{B}_3 = \hat{D}_2$ [both $90^\circ - x$]	✓ R ✓ $\hat{B}_3 = 90^\circ - x$ ✓ S		
	$\therefore EB = ED$ [sides opp equal ∠s]	✓ R	(4)	
				[16]

QUESTION 11

11.1	Construction: Draw diameter SP and join PT $\angle LSR = 90^\circ - \angle RSP$ [tan $\perp$ rad] $\angle STR = 90^\circ - \angle RTP$ [$\angle$'s in a semi circle] $\angle RSP = \angle RTP$ [$\angle$'s in the same segment] $\therefore \angle LSR = \angle STR$	✓ Construction ✓ S ✓ R ✓ S ✓ R ✓ S ✓ R	

OR



Construction: Draw diameter SP and join PR	✓ Construction	
$\angle LSR = 90^\circ - \angle RSP$ [tan $\perp$ rad]	✓ S/R	
$\angle SRP = 90^\circ$ [$\angle$'s in a semi circle]	✓ S ✓ R	
$\angle SPR = 90^\circ - \angle RSP$ [int $\angle$'s of Δ]		
$\angle LSR = \angle SPR$ [both $90^\circ - \angle RSP$]	✓ S	
$\angle SPR = \angle STR$ [$\angle$'s in the same segment]	✓ S/R	
$\therefore \angle LSR = \angle STR$		

OR			
Construction: Join OS and OR $\hat{S}_1 + \hat{S}_2 = 90^\circ$ [tan $\perp$ radius] $\hat{S}_2 = 90^\circ - \hat{S}_1$ $\hat{R}_1 = \hat{S}_2 = 90^\circ - \hat{S}_1$ [$\angle$'s opp equal radii] $\hat{O}_1 = 180^\circ - 2[90^\circ - \hat{S}_1] = 2\hat{S}_1$ [int $\angle$'s of Δ] $\hat{O}_1 = \hat{S}_1$ [at centre = $2 \times \angle$ at circumf] $\therefore \hat{L}\hat{S}\hat{R} = \hat{S}\hat{T}\hat{R}$	✓ Construction ✓ S ✓ R ✓ S/R ✓ S ✓ S ✓ R		
			(6)

Commented [BK8]: Not S but 2T

